Mahdi Mohammadigohari



Machine Learning Theory | Functional Analysis
Kernel and Optimization Geometry
Independent Researcher, Bolzano, Italy
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Research Profile

I develop mathematical foundations for representation learning and deep models using reproducing-kernel Hilbert spaces, variation spaces, operator theory, empirical-process methods, and variational dynamics. My current research program centers on Brownian kernel ladders: recursive function spaces whose geometry can be studied across representation, statistical complexity, approximation, covariance compression, optimization, and mean-field limits. Complementary work develops Koopman- and composition-operator approaches to multi-task generalization. I am particularly interested in rigorous links between adaptive representations, observable quotient geometry, nonsmooth optimization, and certifiable algorithms.

Recursive RKHSs and variation spaces ▪ **Statistical learning theory** ▪ **Gradient flows and mean-field limits** ▪ **Operator-theoretic multi-task learning**

Selected Research Contributions

- **Brownian kernel ladder program.** Developed the foundational recursive integral-RKHS construction and extended it to path-atomic variation spaces, common-covariance geometry, simultaneous approximation, finite certification, and current quotient/relaxed dynamics.
- **Representation selection and Brownian heads.** Derived fixed-representation activation-mass characterizations and same-sample selection bounds through Brownian threshold traces for rectangular and rank-deficient ReLU feature extractors.
- **Optimization geometry.** Isolated how equivalent nodal, increment, and spectral realizations of one Brownian RKHS induce identical or genuinely different optimizer trajectories, including exact Brownian-metric preconditioning and basis dependence of Adam.
- **Operator-theoretic generalization.** Developed Koopman/composition-operator bounds for deep multi-output models in vector-valued Sobolev and Brownian RKHS regimes, together with finite-rank operator learning and transfer guarantees.

Academic and Research Experience

Independent Researcher in Machine Learning Theory Jul 2026–Present
Bolzano, Italy
Developing the Brownian kernel ladder research program, including observable quotient geometry, relaxed multilayer dynamics, adaptive covariance geometry, representation-selection theory, and optimization-aware finite realizations.

Doctoral Researcher Nov 2022–Jun 2026
Faculty of Engineering, Free University of Bozen–Bolzano, Italy
Research on statistical learning theory, multi-task deep learning, operator-theoretic generalization, vector-valued RKHSs, and hierarchical Brownian function spaces. Advisor: Prof. Giuseppe Di Fatta.

Faculty Member / Instructor 2006–2020
Islamic Azad University, Iran
Taught undergraduate mathematics and statistics, including Linear Algebra, Real Analysis, Probability, Statistical Inference, Differential Equations, and Numerical Analysis.

Education

Ph.D. in Computer Science – Machine Learning Theory Nov 2022–Sep 2026
Free University of Bozen–Bolzano, Italy
Dissertation: *Mathematics of Deep Learning: Generalization in Single-Task and Multi-Task Architectures via Operator-Theoretic and Kernel-Based Methods*. Advisor: Prof. Giuseppe Di Fatta.

Graduate Studies in Data Analytics (M.S. curriculum) Sep 2020–May 2022
Rochester Institute of Technology, RIT Dubai GPA: 3.7/4.0
Coursework and Master's Project completed; degree not conferred following a visa interruption. Project: *Energy Consumption Forecasting Using Machine Learning*.

Ph.D. in Mathematics – Functional Analysis Sep 2014–Mar 2018
Islamic Azad University, Mashhad, Iran GPA: 18.7/20
Dissertation: *Operator Inequalities and Operators Preserving Orthogonality*. Advisor: Prof. Maryam Amyari.

M.S. in Pure Mathematics Sep 2003–Feb 2006
Islamic Azad University, Mashhad, Iran

B.S. in Applied Mathematics Sep 1999–Sep 2003
Islamic Azad University, Mashhad, Iran

Publications and Manuscripts

Preprints and submitted manuscripts

[1] **M. Mohammadigohari**. *Variation Brownian Kernel Ladders*. Submitted manuscript, 2026. [arXiv]

[2] **M. Mohammadigohari**, G. Di Fatta, G. Nicosia, and P. M. Pardalos. *Brownian Kernel Ladders*. Submitted manuscript, 2026. [arXiv]

[3] **M. Mohammadigohari**, T. Borsani, and G. Di Fatta. *Operator-Theoretic Generalization Bounds for Multitask Deep Learning*. arXiv preprint, 2026. [arXiv]

[4] **M. Mohammadigohari**. *Common Covariance Geometry and Certification for Brownian Kernel Ladders*. Submitted manuscript, 2026. [arXiv]

[5] **M. Mohammadigohari**. *Sample-Weighted End-to-End Trace-Norm Geometry for Multitask Learning*. Submitted manuscript, 2026. [arXiv]

[6] **M. Mohammadigohari** and N. Mücke. *Brownian Heads for Deep ReLU Representations: Activation Mass and the Cost of Same-Sample Selection*. Submitted manuscript, 2026. [arXiv]

[7] **M. Mohammadigohari** and G. Camps-Valls. *Optimization Geometry of Equivalent Brownian RKHS Representations*. Submitted manuscript, 2026. [arXiv]

Peer-reviewed conference proceedings

[8] **M. Mohammadigohari**, G. Di Fatta, G. Nicosia, and P. M. Pardalos. *On the Koopman-Based Generalization Bounds for Multi-Task Deep Learning*. In Machine Learning, Optimization, and Data Science (LOD 2025), LNCS 16468, Springer, pp. 376–392, 2026. [DOI] [arXiv]

[9] **M. Mohammadigohari**, G. Di Fatta, G. Nicosia, and P. M. Pardalos. *Operator-Based Generalization Bound for Deep Learning: Insights on Multi-Task Learning*. In Machine Learning, Optimization, and Data Science (LOD 2025), LNCS 16468, Springer, pp. 120–137, 2026. [DOI] [arXiv]

Journal articles in functional analysis and applied mathematics

[10] **M. Mohammadi Gohari** and M. Amyari. *The Operator-Valued Parallelism and Norm-Parallelism in Matrices*. Indian Journal of Pure and Applied Mathematics, 51(4):1305–1316, 2020. [DOI]

[11] **M. Mohammadi Gohari** and M. Amyari. *Parallelism in Hilbert K(H)-Modules*. Tbilisi Mathematical Journal, 12(2):67–75, 2019. [DOI] [arXiv]

[12] **M. Mohammadi Gohari** and M. Amyari. *Refinements of the Heinz Inequalities for Operators and Matrices*. Mathematica Slovaca, 68(6):1431–1438, 2018. [DOI]

[13] M. H. Jahandar, M. Elahi Torshizi, H. Farhangfar, and **M. Mohammadi Gohari**. *Using Mathematical Models for the Lactation Curve of Milk and Fat Production in Iranian Holstein Cows*. Indian Journal of Animal Sciences, 82(9):1016–1021, 2012. [DOI]

Manuscripts in preparation

[14] **M. Mohammadigohari** and N. Mücke. *Spectral Bounds for Brownian Kernel Ladders: Covering, Localization, and Fixed-Design Minimax Risk*. In preparation, 2026.

[15] **M. Mohammadigohari** and N. Mücke. *Threshold Geometry and Margin Certificates for Brownian Kernel Ladders*. In preparation, 2026.

Research Grants and Funded Projects

Post-Lauream Research Grant Dec 2025–Jun 2026

Free University of Bozen–Bolzano, Italy
Project: *AI Engineering Suite to Support Agile and Efficient Software Engineering (AI4SWEng)*. Principal Investigator: Prof. Giuseppe Di Fatta. University Research Grant, CUP I53C250000000006.

Learning Fair Latent Representations with Multi-Task Deep Learning Nov–Dec 2025

MTDL Start-Up Fund
Short-term funded project on fairness-aware multi-task representation learning and its theoretical and empirical evaluation.

Technical Expertise

Methods	RKHSs and variation spaces; functional analysis and operator theory; Rademacher/Gaussian complexity; convex analysis; gradient flows; approximation theory; semidefinite optimization.
ML theory	Statistical learning theory; kernel methods; multi-task and representation learning; deep-learning theory; finite and mean-field models.
Software	Python (PyTorch, NumPy, SciPy, scikit-learn, CVXPY), MATLAB, R, LaTeX, and Git; reproducible theorem-aligned experiments.
Languages	Persian (native); English (proficient).